Project: Worked examples
Title: Section properties
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Area, centroid, second moments of area (Ixx, Iyy, Ixy), principal axes and values, elastic and plastic section moduli, radii of gyration, torsion constant and (I-sections) warping constant for a parametric shape, an arbitrary polygon with holes, or a composite of placed parts. The x axis is horizontal (major axis of an I-section), y vertical.

Outline loops are integrated by Green's theorem: A = ½∮(x dy − y dx), Ixx = ∫y² dA, Iyy = ∫x² dA, Ixy = ∫xy dA, transferred to the centroid by the parallel-axis theorem. Principal: tan 2θ = −2Ixy/(Ixx − Iyy). Elastic modulus Ixx/y_max; plastic modulus from the equal-area axis found by bisection, Wpl = |Q_above| + |Q_below|. Torsion: Saint-Venant series for a solid rectangle, πd⁴/32 for circles, Bredt for closed thin-walled, Σbt³/3 (+ fillet term for rolled I) for open sections. Warping (I): Iw = Iyy (h − t_f)²/4.

Ref: Roark's Formulas for Stress and Strain Appendix A, Table A.1; Ch. 10, Table 10.1 — Properties of sections; torsion of solid and thin-walled sections
Ref: AISC Design Guide 9 (Seaburg & Carter) Appendix — Torsional properties of rolled I-shapes
Ref: BS EN 10365 — Hot rolled steel channels, I and H sections: dimensions and masses
Ref: BS EN 10210-2 — Hot finished structural hollow sections: tolerances, dimensions and sectional properties

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
shape	Shape	polygon		drawing: DRG-BG-100 rev C	any
vertices	Polygon vertices (polygon shape)	8 row(s)		drawing: DRG-BG-100 rev C	any
vertices[1]		x = -200 mm; y = -300 mm; loop = 1			
vertices[2]		x = 200 mm; y = -300 mm; loop = 1			
vertices[3]		x = 200 mm; y = 300 mm; loop = 1			
vertices[4]		x = -200 mm; y = 300 mm; loop = 1			
vertices[5]		x = -180 mm; y = -280 mm; loop = 2			

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
vertices[6]		x = 180 mm; y = -280 mm; loop = 2			
vertices[7]		x = 180 mm; y = 280 mm; loop = 2			
vertices[8]		x = -180 mm; y = 280 mm; loop = 2			

3. Assumptions

A1. A polygon outline is simple (no self-intersection); holes lie inside the outline and do not overlap. (Green's theorem integrates the signed loops as given.)
A2. Rolled-section root fillets are quarter circles of radius r; flange tips are square; hollow-section corners are quarter circles of outer radius r_o (default 1.5 t) and inner radius r_i (default 1.0 t). (Published section tables use the same idealisation.)
A3. Thin-walled torsion constants neglect the junction and tip effects except the fillet term kept for rolled I-sections. (Open sections: $\Sigma bt^3/3$; closed: Bredt.)

4. Working

$$[2]\ A = \tfrac{1}{2} \oint (x\,dy - y\,dx)$$ Roark Table A.1

= Green's theorem over 8 vertices in 2 loop(s)
= 384 cm²
Area

$$[3]\ x_c = \int x\,dA / A$$

= Green's theorem over 8 vertices in 2 loop(s)
= 0 mm
Centroid x (from the shape origin)

$$[4]\ y_c = \int y\,dA / A$$

= Green's theorem over 8 vertices in 2 loop(s)
= 0 mm
Centroid y (from the shape origin)

$$[5]\ I_{xx} = \int y^2\,dA - A\,y_c^2$$

= Green's theorem over 8 vertices in 2 loop(s)
= 193150 cm⁴
Second moment of area about the centroidal x axis

$$[6]\ I_{yy} = \int x^2\,dA - A\,x_c^2$$

= Green's theorem over 8 vertices in 2 loop(s)
= 102270 cm⁴
Second moment of area about the centroidal y axis

$$[7]\ I_{xy} = \int xy\,dA - A\,x_c\,y_c$$

= Green's theorem over 8 vertices in 2 loop(s)
= 0 cm⁴
Product of inertia about the centroid

$$[8]\ I_1 = (I_{xx} + I_{yy})/2 + \sqrt{((I_{xx} - I_{yy})/2)^2 + I_{xy}^2}$$

= (193150 cm⁴ + 102270 cm⁴) / 2 + sqrt(((193150 cm⁴ - 102270 cm⁴) / 2)^2 + (0 cm⁴)^2)
= 193150 cm⁴
Major principal second moment

[9] I_2 = (I_xx + I_yy)/2 - sqrt(((I_xx - I_yy)/2)^2 + I_xy^2)
= (193150 cm⁴ + 102270 cm⁴) / 2 - sqrt(((193150 cm⁴ - 102270 cm⁴) / 2)^2 + (0 cm⁴)^2)
= 102270 cm⁴
Minor principal second moment
[10] theta = ½ atan2(-2 I_xy, I_xx - I_yy)
= ½ atan2(0, 90880)
= 0 deg
Angle from the x axis to the major principal axis (anticlockwise)
[11] y_top = y_max - y_c
= Green's theorem over 8 vertices in 2 loop(s)
= 300 mm
Extreme fibre above the centroid
[12] y_bot = y_c - y_min
= Green's theorem over 8 vertices in 2 loop(s)
= 300 mm
Extreme fibre below the centroid
[13] x_ext = max(x_max - x_c, x_c - x_min)
= Green's theorem over 8 vertices in 2 loop(s)
= 200 mm
Extreme fibre from the centroid, x
[14] W_el_x = I_xx / max(y_top, y_bot)
= 193150 cm⁴ / max(300 mm, 300 mm)
= 6438.4 cm³
Elastic section modulus about x (governing fibre)
[15] W_el_y = I_yy / x_ext
= 102270 cm⁴ / 200 mm
= 5113.6 cm³
Elastic section modulus about y
[16] W_pl_x = |Q_above| + |Q_below| about the equal-area axis
= equal-area axis at y = 0 mm
= 7776 cm³
Plastic section modulus about x
[17] W_pl_y = |Q_right| + |Q_left| about the equal-area axis
= equal-area axis at x = 0 mm
= 5856 cm³
Plastic section modulus about y
[18] i_x = sqrt(I_xx / A)
= sqrt(193150 cm⁴ / 384 cm²)
= 22.428 cm
Radius of gyration about x
[19] i_y = sqrt(I_yy / A)
= sqrt(102270 cm⁴ / 384 cm²)
= 16.32 cm
Radius of gyration about y
[20] alpha_pl_x = W_pl_x / W_el_x
= 7776 cm³ / 6438.4 cm³
= 1.2078
Shape factor about x

5. Results

SYMBOL	VALUE	DESCRIPTION
A	384 cm²	Area
x_c	0 mm	Centroid x

SYMBOL	VALUE	DESCRIPTION
y_c	0 mm	Centroid y
I_xx	193150 cm⁴	I_xx
I_yy	102270 cm⁴	I_yy
I_xy	0 cm⁴	I_xy
I_1	193150 cm⁴	I_1 (major principal)
I_2	102270 cm⁴	I_2 (minor principal)
theta	0 deg	Principal axis angle
W_el_x	6438.4 cm³	W_el,x
W_el_y	5113.6 cm³	W_el,y
W_pl_x	7776 cm³	W_pl,x
W_pl_y	5856 cm³	W_pl,y
i_x	22.428 cm	i_x
i_y	16.32 cm	i_y
alpha_pl_x	1.2078	Shape factor, x

6. Checks (governing first)

No pass/fail checks in this calculation.

7. Applicability

WARNING — no-torsion: J: no torsion constant is computed for an arbitrary polygon.

8. Verification

Template struct.section-properties@1.0.0 Content hash 7f7a93ac64011ba4 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0